Polar Coordinates & Polar Graphs

Section 1: Coordinate Conversions (Polar <-> Rectangular)

1 Convert the polar point $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$ to rectangular coordinates (x, y).

a. $(-3\sqrt{3}, 3)$

b. $(3\sqrt{3}, -3)$

c. $(-3, 3\sqrt{3})$

d. $(3, -3\sqrt{3})$

2 Convert the rectangular point $(-4, 4)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $\left(4\sqrt{2}, \frac{3\pi}{4}\right)$

b. $\left(4\sqrt{2}, \frac{7\pi}{4}\right)$

c. $\left(8, \frac{3\pi}{4}\right)$

d. $\left(4\sqrt{2}, \frac{5\pi}{4}\right)$

3 Which polar point represents the same location as $(r, \theta) = \left(3, \frac{\pi}{3}\right)$?

a. $\left(-3, \frac{4\pi}{3}\right)$

b. $\left(-3, \frac{\pi}{3}\right)$

c. $\left(3, -\frac{\pi}{3}\right)$

d. $\left(-3, \frac{2\pi}{3}\right)$

Section 2: Converting Equations between Polar and Cartesian

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4 Convert the polar equation $r = 6\cos\theta$ into rectangular coordinates.

a. $(x-3)^2 + y^2 = 9$

b. $x^2 + (y-3)^2 = 9$

c. $x^2 + y^2 = 6x$

d. $x^2 + y^2 = 36$

5 Convert the rectangular equation $x^2 + y^2 = 16$ to polar form.

a. $r = 4$

b. $r = 16$

c. $r = 4\cos\theta$

d. $r = 16\sin\theta$

6 Convert $r = 4\csc\theta$ to rectangular coordinates.

a. $y = 4$

b. $x = 4$

c. $y = 4x$

d. $x^2 + y^2 = 16$

Section 3: Rose Curves and Symmetry

7 How many petals does the rose curve $r = 5\cos(4\theta)$ have?

a. 8

b. 4

c. 16

d. 2

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8 How many petals does the rose curve $r = 3\sin(5\theta)$ have?

a. 5

b. 10

c. 15

d. 3

9 What is the maximum distance from the pole (origin) to the curve $r = 4\sin(3\theta)$?

a. 4

b. 12

c. 3

d. 8

Section 4: Limaçons and Special Polar Curves

10 Classify the polar curve $r = 2 + 2\cos\theta$.

a. Cardioid (heart-shaped limaçon)

b. Limaçon with an inner loop

c. Dimpled limaçon

d. Convex limaçon

11 Which polar equation produces a limaçon with an inner loop?

a. $r = 2 + 4\cos\theta$

b. $r = 4 + 2\cos\theta$

c. $r = 3 + 3\sin\theta$

d. $r = 5 + 4\cos\theta$

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12 At what angles $\theta \in [0, 2\pi)$ does the curve $r = 2 + 4\cos\theta$ pass through the pole ($r = 0$)?

a. $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$

b. $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$

c. $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$

d. $\theta = \pi$

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